

AI Agent CRM 完整架構設計方案

設計原則：一層嚴謹，一層彈性，兩層唔好撈埋

1. 核心架構分層

層	職責	類型
用戶層	自然語言輸入 / 輸出呈現	彈性
Agent 層	意圖分類、決策路由、對話管理	彈性
Tools 層	結構化 I/O，Pydantic Schema，統一介面	彈性
嚴謹層	狀態機、權限、Validation、審計	嚴謹 ★
數據層	PostgreSQL + Audit Log，所有寫操作 immutable	嚴謹 ★

2. Tool 執行流程

- LLM 分析用戶輸入，提出 action（自然語言）
- Structured Output（Pydantic Schema）— 唔係直接執行
- Tool Function 接收結構化 input
- 嚴謹層校驗：狀態機、權限、Validation
 - Pass → 寫 DB + Audit Log → return structured result
 - Fail → return 具體 error → LLM 重新組織回覆用戶

3. Tool Definitions

Lead Tools

create_lead(name, email, company?, source?)

→ Validation: name non-empty, email format | 業務: duplicate company+email 阻擋 | 寫: DB insert, status="new" | Audit: action="lead.created"

update_lead(lead_id, **fields)

→ Validation: lead exists, state allows update | 業務: status transition rules | 寫: DB update | Audit: action="lead.updated", changes=before/after

qualify_lead(lead_id)

→ Precondition: lead.status IN (new, contact_made) | 業務: at least one contact required | 寫: status="qualified", qualified_at=now | Audit: action="lead.qualified"

assign_lead(lead_id, owner_id)

→ Validation: owner exists, owner.role=sales | 業務: reassignment audit | 寫: owner_id update | Audit: action="lead.assigned"

get_my_leads(status?, page?)

→ Row-level filter always applied: WHERE owner_id = :current_user | 分頁返回

Opportunity Tools

create_opportunity(lead_id, title, value?)

→ Precondition: lead.status = qualified | Validation: duplicate opp per lead check | 寫: DB insert, stage="qualified" | Audit: action="opportunity.created"

update_stage(opp_id, new_stage, reason?)

→ State machine enforced | Validation: transition allowed | 寫: DB update, stage_changed_at=now | Audit: action="opportunity.stage_changed"
→ Blocked: return specific error → LLM rephrases

add_task(opp_id?, lead_id?, title, due_date?)

→ Validation: either opp_id or lead_id required | 寫: DB insert | Audit: action="task.created"

send_email(to, subject, body)

→ Always draft mode | 寫: DB insert, status="draft" | Audit: action="email.drafted"
→ Requires explicit confirmation before send

get_opportunity_history(opp_id)

→ Immutable audit trail | SELECT FROM audit_log WHERE entity_type='opportunity' AND entity_id=opp_id

Contact Tools

create_contact(lead_id, name, email?, phone?)

→ Validation: lead exists | 業務: link to company | 寫: DB insert | Audit: action="contact.created"

link_contact_to_opportunity(contact_id, opp_id)

→ Validation: contact exists, same company check | 業務: duplicate link check | 寫: contacts.opp_id update | Audit: action="contact.linked_to_opportunity"

4. State Machines

Lead State Machine

new → contact_made → qualified → (create opportunity)
■ unqualified (blocked)

Transition Rules:

new → contact_made: add_contact() called

contact_made → qualified: qualify_lead() + at least one contact

qualified → unqualified: BLOCKED

Opportunity State Machine

qualified → discovery → proposal → negotiation → won ✓
■ lost ✗

Transition Rules:

qualified → discovery: add_contact() linked to this opp

discovery → proposal: send_quote() called, quote prepared

proposal → negotiation: quote sent to contact

negotiation → won: proposal_sent=true, value>0, (value<=approval_limit OR approved=true)

negotiation → lost: close_lost() + reason_code required — BLOCKED if no reason

5. Database Schema

Core Tables:

users – id (PK), email, name, role, team_id (FK → teams), created_at
leads – id (PK), name, email, company, source, status, owner_id (FK), qualified_at, created_at/updated_at
opportunities – id (PK), title, value, stage, lead_id (FK), owner_id (FK), stage_changed_at, expected_close, lost_reason, created_at/updated_at
tasks – id (PK), title, description, status, opp_id (FK nullable), lead_id (FK nullable), assigned_to (FK), due_date, completed_at, created_at
contacts – id (PK), name, email/phone/role, company_id (FK → leads), opp_id (FK nullable), is_primary, created_at
emails – id (PK), opp_id (FK nullable), lead_id (FK nullable), sender_id (FK), direction, subject/body_preview, status, sent_at, created_at
audit_log – id (PK), entity_type, entity_id, action, actor_id (FK), actor_type, changes (JSONB), reason, created_at

⚠ NO UPDATE/DELETE allowed — DB trigger enforces immutability

Relationships:

users ■■teams
■■■ leads (owner_id FK)
■■■contacts (company_id FK)
■■■ opportunities (lead_id FK, owner_id FK)
■■■ tasks (opp_id FK, assigned_to FK)
■■■ emails (opp_id FK, sender_id FK)
■■■ audit_log (immutable)

6. Row-Level Security

Role	SELECT	MUTATION
admin	All records in their team	All records in their team
manager	Own + reports records	Own + reports records
sales	owner_id = current_user.id ONLY	owner_id = current_user.id ONLY
system (agent)	ALL via audit context	mutation via Tools only

Special Restrictions:

- lost_reason: Manager+ only. Sales cannot see other users' lost reasons
- audit_log writes: Only Tools / system. No direct INSERT allowed
- audit_log deletes: PROHIBITED. DB trigger blocks UPDATE/DELETE
- email auto-send: Always draft mode by default. Requires explicit confirmation
- high-value deal approval: Value > manager_approval_limit requires explicit approval task